

Introduction to Control Systems (ME2801)

Homework #1

The textbook exercises, designated by with the word “Nise”, are from “Problems” section at the end of the chapter in “Control System Engineering” by Norman Nise, 7th edition.

Laplace Transform of Functions

1) Nise 2.2

Using the Laplace transform pairs and theorems, find the Laplace transform ($F(s)$) of the following time functions:

- a) $f(t) = e^{-at} \sin(\omega t) u(t)$
- b) $f(t) = e^{-at} \cos(\omega t) u(t)$
- c) $f(t) = t^3 u(t)$

2) Given the following time functions ($f(t)$), find the Laplace transform($F(s)$) of each function.

- d) $f(t) = 1 + 2t$
- e) $f(t) = 3 + 7t + t^2 + \delta(t)$
- f) $f(t) = e^{-t} + 2e^{-2t} + te^{-3t}$
- g) $f(t) = 3 \cos(6t)$
- h) $f(t) = 2 \cos(2t) + e^{-t} \sin(2t)$

3) For each of the following time functions

- a) $f(t) = 10.0(e^{-0.5(t)} \sin(2\pi t))$
- b) $g(t) = 10.0(e^{-0.75(t)} \sin(2\pi t))$
- c) $h(t) = 10.0(e^{-0.5(t)} \sin(4\pi t))$

3.1) Find the Laplace transform, i.e $F(s)$, $G(s)$ and $H(s)$

3.2) Plot the time-domain functions using MATLAB

- Use the `linspace` command to create a time vector from $t=0$ to $t=10$ seconds with 1000 points
- Submit a single, well-formatted figure that shows all three functions on the same axes and includes the following properties
 - Provide a title and axes labels with units (the y-axis label can be “Time Response [n/a]”)
 - Use different line styles (combination of colors, type (solid, dashed, etc.) and line weight)
 - Include a legend
 - Turn graph lines on.

You are not required to submit the MATLAB script – just the figure.

Inverse Laplace Transform of Functions

4) Given the following Laplace transforms ($F(s)$), find corresponding time functions ($f(t)$). Use the MATLAB `residue` function for the partial fraction expansion.

You do not need to turn in your MATLAB scripts.

a) $F(s) = \frac{2}{s(s+2)}$

b) $F(s) = \frac{10}{s(s+1)(s+10)}$